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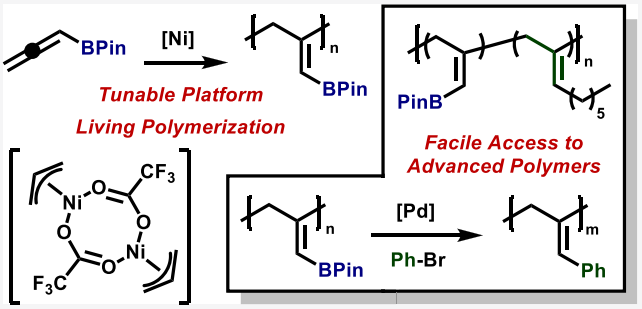
Vinyl-Addition Polymerizations of Borylated Allenes

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ABSTRACT: Borylated motifs hold immense potential for functional soft materials, as they can facilitate a range of reaction manifolds (e.g., analyte binding or cross-coupling). Traditional boron-containing materials are predominantly composed of either aryl boronates or main-chain embedded vinylic boronates. Progress toward expanding the architectural (and, by extension, functional) scope of borylated polymers remains slightly limited. To address this, we report the vinyl-addition polymerization of an allenylboronic acid pinacol ester. The base metal catalyst, $[(\eta 3\text{-}C_3H_5)Ni(OC(O)CF_3)]_2$, was found to readily polymerize this allene monomer with good regioselectivity and molecular weight control. Additionally, the allenylboronate monomer could be copolymerized with other allenes (e.g., 1,2-nonadiene or β-pinadiene) to access an array of borylated materials. The borylated polyallenes could also be chemically tailored under mild conditions (e.g., using Pd-catalyzed cross-coupling), which suggests that they could serve as general platforms for accessing diverse architectures.



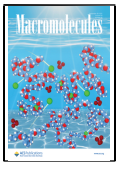
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INTRODUCTION

Soft materials that incorporate main group elements often exhibit unique physical or chemical properties, and synthetic access to these scaffolds has long been a focus in polymer science. Borylated polymers are particularly attractive targets, as boronates can exhibit a broad range of reactivity manifolds.[1−14] This fact is well established for small molecule chemistry, where boron-containing substrates are key components of C−C bond-forming reactions (e.g., Suzuki−Miyaura,[4−6] Matteson,[8−10] Brown,[11,12] or Zweifel[13] chemistry) that have profoundly transformed the art of organic synthesis. Additionally, the Lewis acidity of boron is routinely exploited for the development of chemosensors.[14] Interfacing borylated motifs with polymers is increasingly a key design element in the development of advanced materials,[15−21] and recent progress has showcased boron-containing polymers in a variety of applications.[22−38]

While these precedents demonstrate the versatility of unifying boron chemistry with polymer science, there are lingering obstacles to the preparation of borylated materials. For example, the synthetic scope is largely limited to polymers derived from aromatic boronates (although materials bearing aliphatic boronates were recently prepared using radical[24,25] or metal-catalyzed polymerizations[26]).[15−21] Jäkle reported early examples of preparing well-defined borylated polystyrenes either through a substituent exchange reaction[15] or through ATRP.[16] Summerlin would later elaborate this work by developing RAFT polymerizations of styrene monomers that were functionalized with boronic esters (Figure 1A).[17] Polyacrylamides[18] and polylactides[35] decorated with aryl boronate derivatives have also been prepared through radical polymerization or Miyaura-borylation, respectively (Figure 1B). Controlled access to polymers exhibiting vinylic sp^2 C−B bonds, however, is slightly more limited in scope. Initial efforts by Chujo detailed the hydroboration polymerization of bifunctional boranes across a π-conjugated enediyne synthon to furnish fully conjugated organoborane polymers.[19] Qin elaborated further on this methodology by employing *trans*-enediynes as coupling partners, yielding borylated polyacetylenes (Figure 1C).[33] Notably, these strategies afforded vinylic boronates within the polymer scaffold (which enabled tuning of optoelectronic properties).[21] However, current methods for incorporating vinyl boronates into polymers impose certain restrictions. For example, such materials are limited to alternating copolymers, and reported examples feature low molecular weights (<13,000 Da). Given our recent interest in exploring the interface of small molecule chemistry and borylated polymers,[28] we reasoned that these platforms are not well suited for chemical tailoring (as cross-coupling through the boronate in the main chain could destroy the polymer). Thus, we envisioned that new approaches for generating polymers that are decorated with chemically addressable vinyl boronates would represent a significant advancement.

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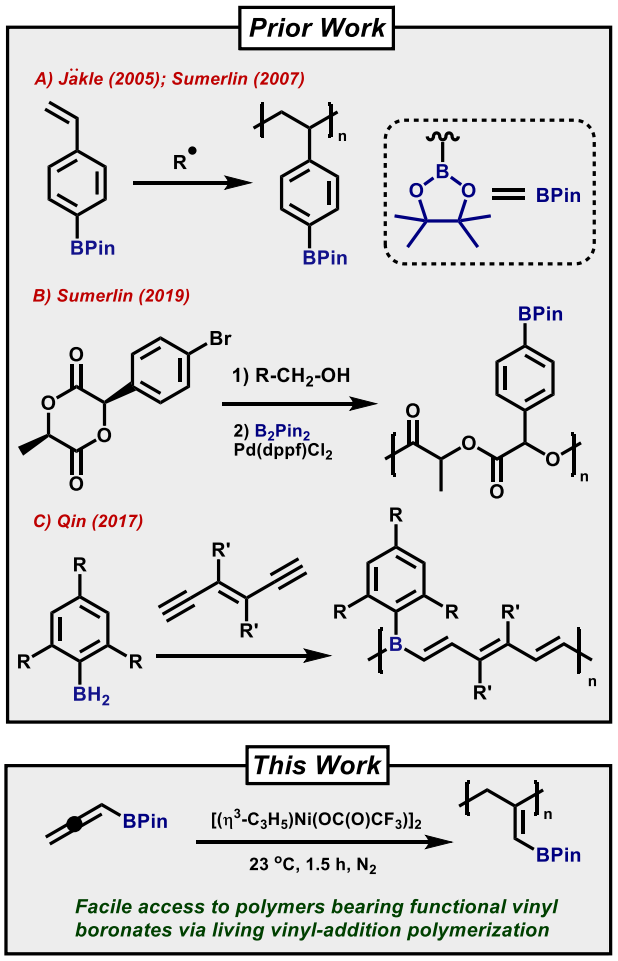


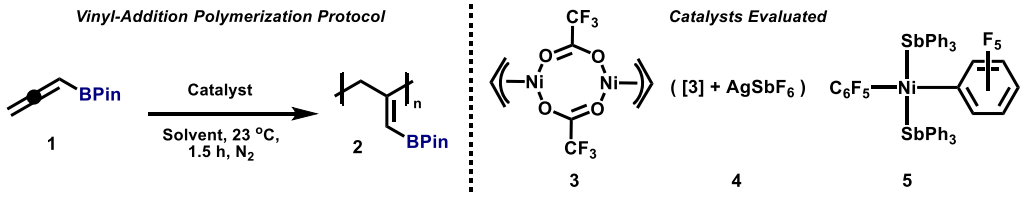
Figure 1. Selected examples of borylated polymers with aryl (A,B) or vinylic (C) C−B motifs.

We surmised that vinyl-addition polymerization of borylated allenes could represent an unexplored route to address this challenge. Endo has reported that $[(\eta^3\text{-}C_3H_5)Ni(OC(O)\text{-}CF_3)]_2$ catalyzes the living vinyl-addition polymerizations of allenes decorated with an array of functional groups,[39−41] and sterically encumbered monomers often exhibit excellent regioselectivity for 2,3-insertions. Given this precedent, we hypothesized that bulky allenyl boronates (e.g., pinacol esters) would exhibit similar regioselectivity during vinyl-addition polymerizations to afford $C(sp^2)_{vinyl}$−B motifs in the side chain of the repeating unit (Figure 1). Realization of this process would allow for the generation of polymers bearing vinylic boronates that could permit both (i) tailored copolymer generation and (ii) post-polymerization modifications of the pendant vinyl boronates for further structural diversification.

RESULTS AND DISCUSSION

Polymerization Optimization. Allenylboronic acid pinacol ester **1** was selected for our preliminary investigations, given that (i) the pinacol ester should be sufficiently encumbering to promote the desired 2,3-regioselectivity and (ii) **1** can be readily synthesized on multigram scales.[42] Our polymerization studies commenced by evaluating the efficacy of common vinyl-addition polymerization catalysts (Table 1, entries 1−3). We initially focused on $[(\eta^3\text{-}C_3H_5)Ni(OC(O)\text{-}CF_3)]_2$ (**3**; prepared *in situ*; see Supporting Information), given that it has been shown to polymerize allenes with excellent regioselectivity and functional group compatibility.[39−41] ^{1}H NMR spectroscopy (toluene-d_8) revealed 99% consumption of **1** ($[\mathbf{1}]_o$ = 0.5 M) in the presence of **3** after 1.5 h at 23 °C. The mixture was then filtered through basic alumina and concentrated to dryness to afford polymer **2** as a colorless solid (92% yield). Analysis by gel permeation chromatography (GPC) revealed that **2** exhibited a number average molecular weight (M_n) that was in good agreement with the targeted value (cf. 27,260 vs 30,000 Da, respectively). ^{1}H NMR spectroscopic analysis of **2** was consistent with the expected product regioselectivity to afford vinyl boronates in the repeating unit, as we observed a broad vinyl resonance (δ 5.26 ppm) that integrated to 1 H (relative to the pinacol methyl groups). The expected allylic methylene proton resonances (δ 2.5−3.5 ppm) and pinacol methyl proton resonances (δ 1.16 ppm) were also observed (Figure 2). Furthermore, DEPTQ NMR spectroscopic analysis of **2** was

Table 1. Optimization for Vinyl-Addition Polymerization of Allene Monomer 1[a]



entry	catalyst	solvent	% yield	target M_n (Da)	M_n (Da)[b]	$Đ$[c]
1	**3**	toluene	92	30,000	27,260	1.53
2	**4**	toluene	56	30,000	70,600	1.20
3	**5**	toluene	8	30,000	N.D.[d]	N.D.[d]
4	**3**	THF	82	30,000	28,000	2.08
5	**3**	Et_2O	90	30,000	35,180	1.49
6	**3**	$CDCl_3$	80	30,000	34,420	1.52
7	**3**	DCM	90	30,000	29,450	1.38
8[e]	**3**	DCM	96	30,000	39,970	1.51

[a]Standard polymerization conditions (see Supporting Information): **1** (0.6 mmol, $[\mathbf{1}]_o$ = 0.5 M), catalyst (0.0033 mmol Ni), 23 °C, 1.5 h. [b]Determined by GPC with triple detection. [c]Calculated from M_w/M_n. [d]Isolated material insoluble in THF solvent. [e]Polymerization conducted using **1** (2.0 g; 12.05 mmol).

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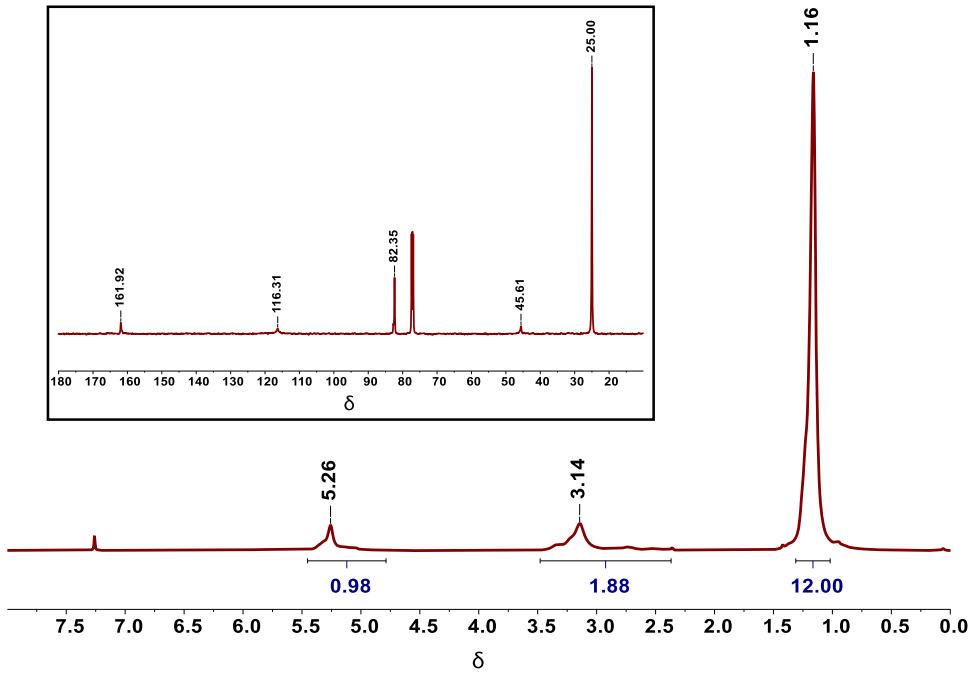


Figure 2. ^{1}H and ^{13}C (inset) NMR spectra of polymer **2** in $CDCl_3$.

consistent with the proposed repeating unit (Figure S17). Specifically, one vinyl resonance (δ 116.1 ppm) was associated with a $C(sp^2)$−H group, while the other vinyl resonance (δ 161.7 ppm) was associated with a quaternary center. If the polymerization proceeded by 1,2-insertion, we would expect that one vinyl resonance would be associated with a $C(sp^2)$−H_2 center. As such, our combined NMR studies were consistent with the expected 2,3-polymerization of **1**.

Closer inspection of the broad resonances in the ^{1}H NMR spectrum revealed major and minor resonances within each range. Integration of these ranges revealed a consistent ratio of 1 vinyl proton to 2 methylene protons in each set of resonances (Figures S19 and S20); thus, we did not attribute these minor resonances to products of 1,2-insertions. Instead, we hypothesized that these minor resonances (δ 2.49−2.72 and 5.04−5.13 ppm) could have originated from (*Z*)-insertion of the allene during the polymerization. Importantly, the corresponding ^{11}B NMR spectrum displayed a broad resonance (δ 24.25 ppm; Figure S21), which indicated retention of the boronate ester within the isolated polymer.[43]

Further solid-state analyses of the isolated polymer were conducted to assess some of the bulk material properties of product **2**. Thermogravimetric analysis of homopolymer **2** revealed a decomposition onset temperature of 255 °C (Figure S8), and dynamic mechanical analysis (DMA) of a powder sample revealed T_g = 147 °C (determined using $\tan(\delta)_{max}$; Figure S9). This value was further corroborated by differential scanning calorimetry, which revealed a transition at 148 °C (Figure S10).

Having demonstrated successful polymerization of **1**, we sought to briefly survey the efficacy of other vinyl-addition polymerization catalysts. We focused on *trans*-$(C_6F_5)_2Ni(SbPh_3)_2$ (**5**) and a cationic Ni-allyl catalyst that is highly active in the vinyl-addition polymerizations of olefins (**4**; generated by activating **3** with $AgSbF_6$).[44] Catalyst **4** afforded the desired product in an acceptable yield (52%; Table 1, entry 2), but GPC analysis revealed a bimodal product distribution (Figure S2). Catalyst **5** displayed low reactivity toward the polymerization of **1**, with only 8% yield of the polymer isolated after 1.5 h (Table 1, entry 3). Moreover, addition of this reaction mixture to MeOH precipitated a solid that was sparingly soluble in common organic solvents (which precluded further analyses).

Given these outcomes, we elected to use **3** for our subsequent optimization studies. We initially surveyed various solvents to identify their impact on the target polymerization (Table 1, entries 4−8). Tetrahydrofuran (THF) afforded polymer **2** in yield similar to toluene (Table 1, entry 4), but GPC analysis of the THF-derived product revealed a bimodal distribution of products (Figure S3). Diethyl ether provided the product in high isolated yield (Table 1, entry 5), but the observed M_n greatly exceeded the target (cf. 35,180 vs 30,000 Da, respectively). Dichloromethane (DCM), however, afforded a low dispersity material in high yield (Table 1, entry 7), and the observed M_n was in excellent agreement with the target value (cf. 29,450 vs 30,000 Da). As such, DCM was selected as our solvent of choice for subsequent studies.[45]

With our selected polymerization conditions in hand, we successfully prepared materials exhibiting a range of molecular weights (M_n = 24,070−130,800 Da; $Đ$ = 1.42−1.70) in modest to good yields (69−87%; Table S1). We note that dilute conditions ($[\mathbf{1}]_o$ = 0.1 M) were necessary to access higher M_n materials with good control over molecular weight and dispersity. For example, when targeting a material with M_n = 100,000 Da under our standard concentration ($[\mathbf{1}]_o$ = 0.5 M), we obtained a polymer with a measured M_n that far exceeded the target (M_n = 130,800 Da; Figure S5). Conducting the same polymerization using $[\mathbf{1}]_o$ = 0.1 M, however, afforded a material with a measured M_n that more closely matched the theoretical value (M_n = 90,350 Da; Figure S6).

Living Character Studies. Having demonstrated successful vinyl-addition polymerizations of **1**, we sought to determine if the reaction exhibited living character. We monitored M_n as a function of monomer conversion, which revealed a linear

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correlation consistent with a living polymerization (Figure 3A). Additionally, the dispersity values narrowed considerably with

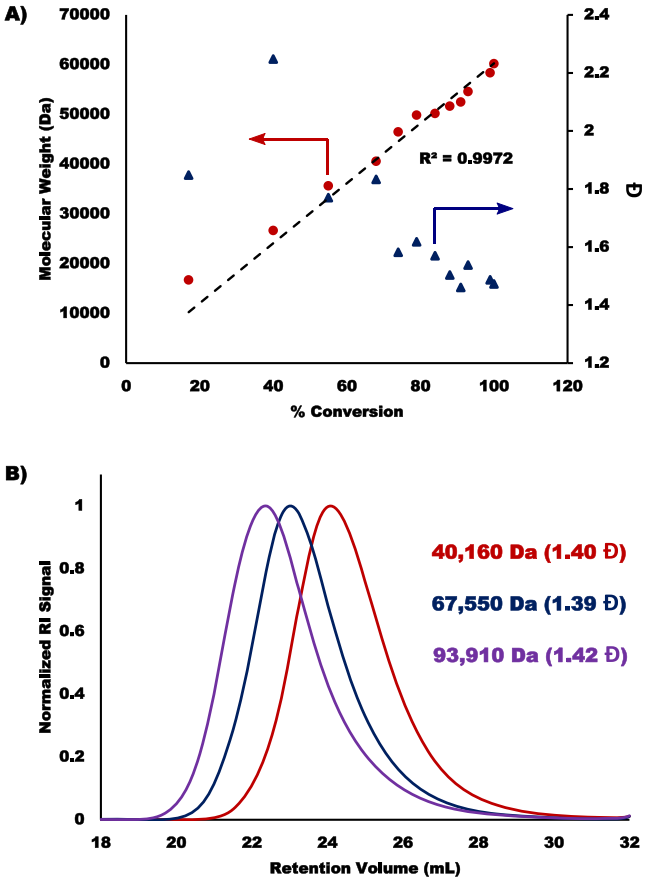


Figure 3. Evaluation of the living character of the polymerization of **1**. (A) Plot of molecular weight as a function of monomer conversion and (B) GPC traces for a chain extension experiment.

increasing monomer conversion, further implicating the living nature of this polymerization.[46] To further explore the nature of the polymerization, we also conducted a chain extension experiment (Figure 3B; see Supporting Information). Briefly, the polymerization of **1** was monitored by ^{1}H NMR spectroscopy, until complete consumption of the monomer was observed, after which an aliquot was removed and analyzed by GPC. A second portion of **1** was added to the polymerization, and the reaction was again monitored by ^{1}H NMR spectroscopy. Upon consumption of the added monomer, an aliquot was removed for GPC analysis. This sequence was repeated an additional time, and the GPC data were consistent with successful chain extensions (Table S3). Lastly, a low M_n sample of **2** was prepared (M_n = 4865 Da, 1.45 $Đ$; Figure S7) for end-group analysis *via* ^{1}H NMR spectroscopy (Figure S18).[48] Using the allyl resonances as diagnostic handles, the calculated M_n (M_n = 3363 Da) was in good agreement with the value obtained from GPC (which suggested high end-group fidelity).These experiments suggest that the polymerization has a living character, but the dispersity ($Đ$ > 1.47) of the resultant polymers indicates that molecular weight control could be slightly limited (Tables 1 and S2).[47]

Block Copolymerizations with Additional Allenes. Because the polymerization of allene **1** was shown to exhibit a living character, efforts turned toward the generation of block copolymers. Copolymerizations of **1** with alkyl allenes struck us as an enticing target,[39−41,49,50] as the resultant materials could be considered borylated congeners of polyolefins. We selected 1,2-nonadiene as the comonomer for a preliminary study, given that this allene has recently been reported to undergo vinyl-addition polymerization under the action of **3** (Figure 4).[49] A block copolymerization was conducted under

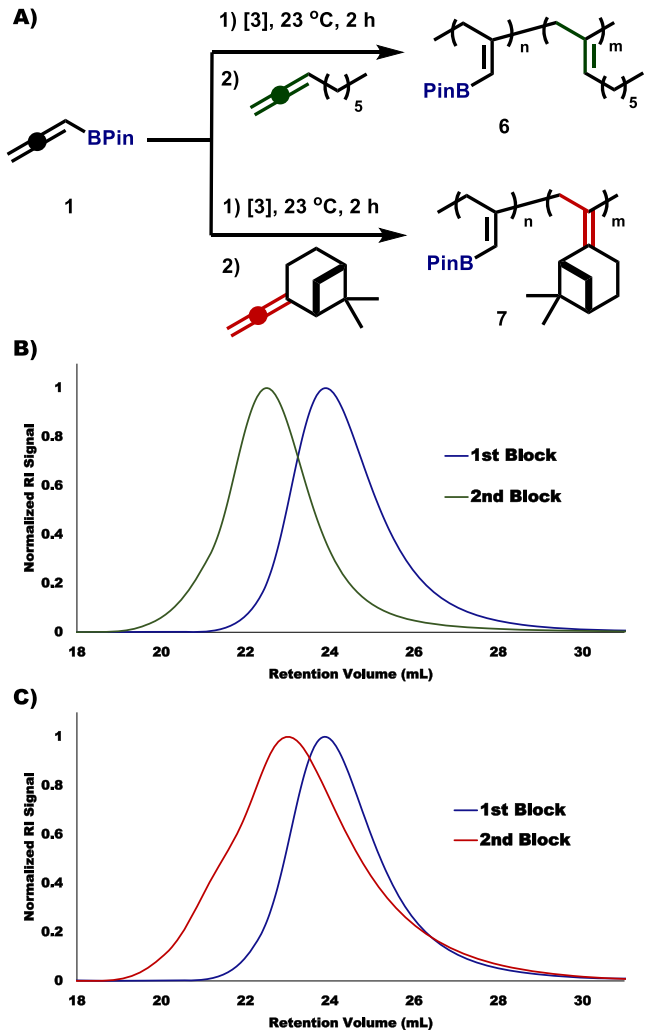


Figure 4. Syntheses of block copolymers. (A) Block copolymerization of monomer **1** with 1,2-nonadiene or β-pinadiene. GPC data for the preparation of **6** (B) and **7** (C).

conditions that targeted an ultimate M_n = 100,000 Da (50,000 Da per block; Figure 4A; see Supporting Information). Briefly, **1** was first polymerized under the action of **3**. Upon complete consumption of the monomer (as determined by NMR spectroscopy), 1,2-nonadiene was added to the polymerization mixture. The final block copolymer, **6**, was precipitated from MeOH after 2 h (46% yield; Table 2). Importantly, ^{1}H NMR spectroscopic analysis confirmed the successful incorporation of **1** into **6** (31 vs 43% predicted; *vide infra*). GPC analysis of **6** revealed that M_n was in modest agreement with the expected value (Figure 4B; M_n = 80,510 vs 100,000 Da predicted; $Đ$ = 1.43), and thermogravimetric analysis of the freshly precipitated material revealed that **6** exhibited good thermal stability (cf. T_d = 331 °C for **6** vs 255 °C for **2**; Figure S14). Unfortunately, the waxy nature of **6** precluded further

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Table 2. Tabulated Data for Block Copolymers[a]

product	initial M_n (Da)[b]	initial $Đ$[c]	final M_n (Da)[b]	final $Đ$[c]	isolated yield (mg)	% 1 theo.[d]	% 1 actual[e]
6	37,360	1.30	80,510	1.43	275.2	43	31
7	35,330	1.34	66,460	1.77	424.5	49	37

[a]Block copolymerization monomer amounts used: 1 (300 mg) and comonomer (300 mg). See Supporting Information for experimental details. [b]Determined by GPC with triple detection. [c]Calculated from M_w/M_n. [d]Theoretical % monomer 1 incorporation calculated from the feed ratio. [e]Actual % monomer 1 incorporation determined by ^{1}H NMR spectroscopy.

evaluation of thermomechanical properties using DMA. Given that poly(1,2-nonadiene) has a degradation onset at approximately 325 °C,[49] incorporation of this motif into 6 could be the origin of the enhanced thermal stability relative to homopolymer 2.

We also prepared a block copolymer of 1 and a biomass-derived monomer to access a material that could be considered a novel analogue of cyclic olefin copolymers. We selected the terpene-derived allene, β-pinadiene, for our efforts (as this monomer has recently been shown to undergo living vinyl-addition polymerizations with 3[50]). Copolymerization of 1 and β-pinadiene was conducted in a manner similar to the procedure described above, and we ultimately isolated block copolymer 7 as a faint-yellow powder (71% yield; Figure 4C). Moderate incorporation of monomer 1 was determined by ^{1}H NMR spectroscopic analysis (37 vs 49% predicted). GPC analysis of 7 revealed a lower M_n than expected (cf. 66,460 vs 100,000 Da, respectively) and a slightly broader dispersity relative to block copolymer 6 (Table 2). The thermal stability of freshly precipitated 7 increased relative to homopolymer 2 (cf. T_d = 273 vs 255 °C; Figure S15), and DMA of a powder sample revealed a higher T_g value as well (cf. 161 vs 147 °C; Figure S16). As in the case of product 6, the physical properties of 7 seemed to be dominated by the comonomer.

We were intrigued by the fact that incorporation of 1 into both block copolymers was lower than expected, even though 1 was fully consumed during both reactions. This fact, combined with modest isolated yields for both 6 and 7, suggested to us that incomplete chain extension could explain our observations. To probe this, we conducted a synthesis of 6 as described above, precipitated the solid polymer from MeOH, and collected the supernatant. Concentration of this supernatant to dryness afforded a colorless powder, and ^{1}H NMR spectral analysis revealed that this solid was homopolymer 2 (which is highly soluble in MeOH). Similarly, 2 was isolated from the supernatant following precipitation of freshly prepared 7. In both cases, the recovered mass of all polymeric products exceeded 74%. Interestingly, switching the order of monomer addition (i.e., 1,2-nonadiene or β-pinadiene was polymerized before adding 1 to the reaction) also resulted in the formation of 2 with a minimal impact on the total recovered mass of polymeric products (see Table S4, Figures S30 and S31). These data, along with the increase in Đ that we observed upon chain extension (Figure 4), suggest that chain transfer of the active polymerization catalyst could be operative. The mechanism (and catalyst speciation) that gives rise to this putative activity is currently being investigated.

Post-polymerization Modification. Given that 2 is decorated with vinyl boronates, we also explored methods for chemically modifying the polymer to demonstrate the synthetic versatility of these motifs. We chose to investigate Suzuki–Miyaura cross-coupling, given the preponderance of recent literature regarding the interfacing of this synthetic method with material science.[28,35,51,52] Polymer 2 exhibits some homology to polyolefins, and we envisaged that successfully merging cross-coupling with our new polymer platform may enable rapid access to diversified, functional polyolefin products. As a simple demonstration of this approach, the cross-coupling of 2 with phenyl bromide was evaluated (Figure 5). The coupled product, 8, was obtained as

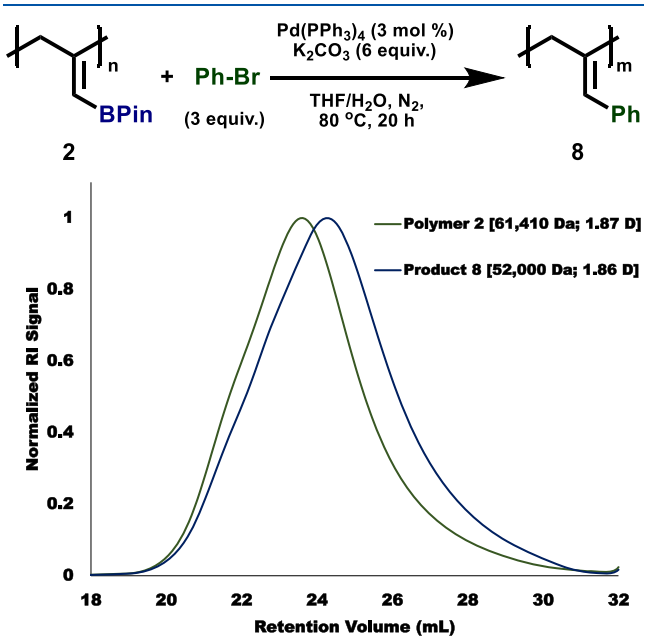


Figure 5. Pd-catalyzed Suzuki–Miyaura cross-coupling of polymer 2.

a solid in high conversion (96%) and excellent yield (93%), and ^{1}H NMR spectroscopic analysis was consistent with the expected product (Figure S32).[39] Gratifyingly, GPC analysis of 8 indicated that the cross-coupling proceeded with high fidelity and minimal changes to overall material dispersity (cf. $Đ$ = 1.86 vs 1.87 for 8 and 2, respectively; Figure 5). The success of this protocol suggests that non-aryl sp^2 C–B motifs could become useful linchpins that engender polymeric scaffolds with intrinsic tunability.

CONCLUSIONS

Borylated polymers that bear arene and vinylic sp^2 C–B motifs have demonstrated considerable versatility, underscoring the importance of pursuing new routes to access these scaffolds. We have detailed the vinyl-addition polymerization of the borylated allene monomer 1 to generate a polymer that is decorated with vinyl boronates. The polymerization exhibited living characteristics, and block copolymers of 1 and other allenes (i.e., 1,2-nonadiene and β-pinadiene) were successfully prepared. Lastly, we have shown that simple Suzuki–Miyaura cross-coupling is possible with polymer 2 to afford functional polyolefin surrogates in a facile manner. We envision that our

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work will serve as an inspiration for the development of other tunable, borylated materials.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.macromol.1c01253.

Additional experimental procedures, characterization data, and results (PDF)

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

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Notes

The authors declare no competing financial interest.

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